

# ELI — Architecture Ascii

ELI v2.2.6

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## Contents

**Blueprint — ELI MKXI Full Architecture (ASCII)**

A. Full system — layered block diagram . . . . .

B. The 15 agents . . . . .

C. Module tree (LOC · key files · role) . . . . .

D. On-disk data layout . . . . .

E. The flow in one line . . . . .

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3

3

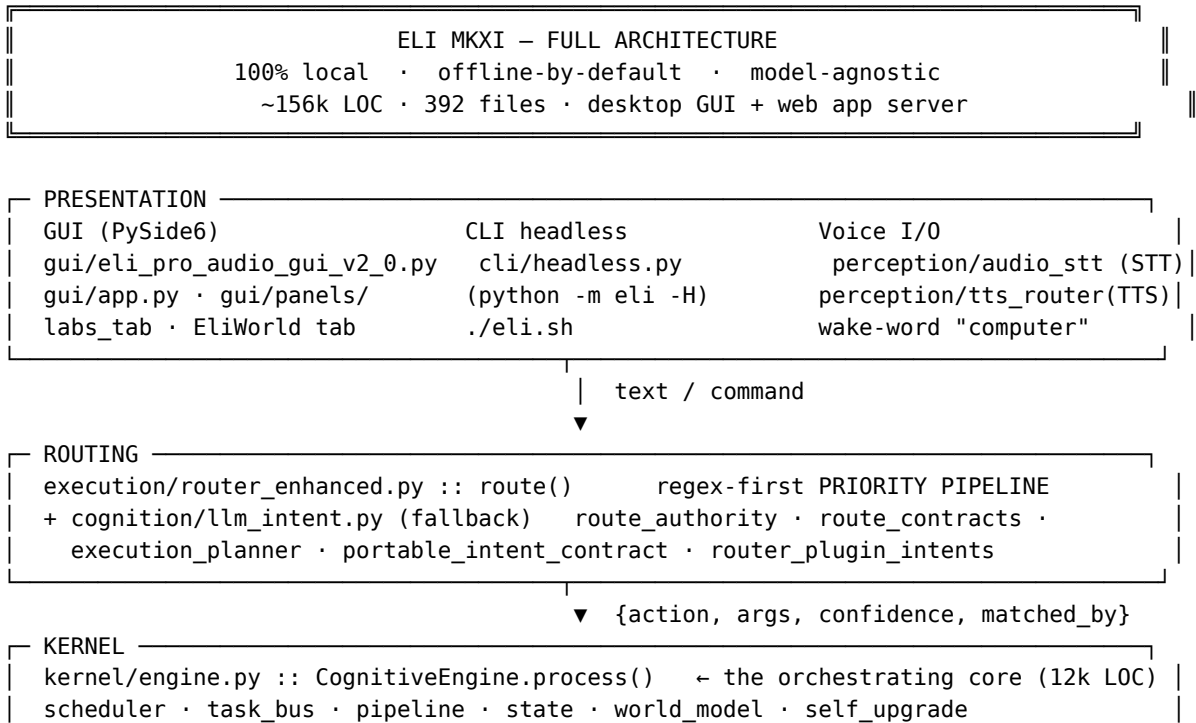
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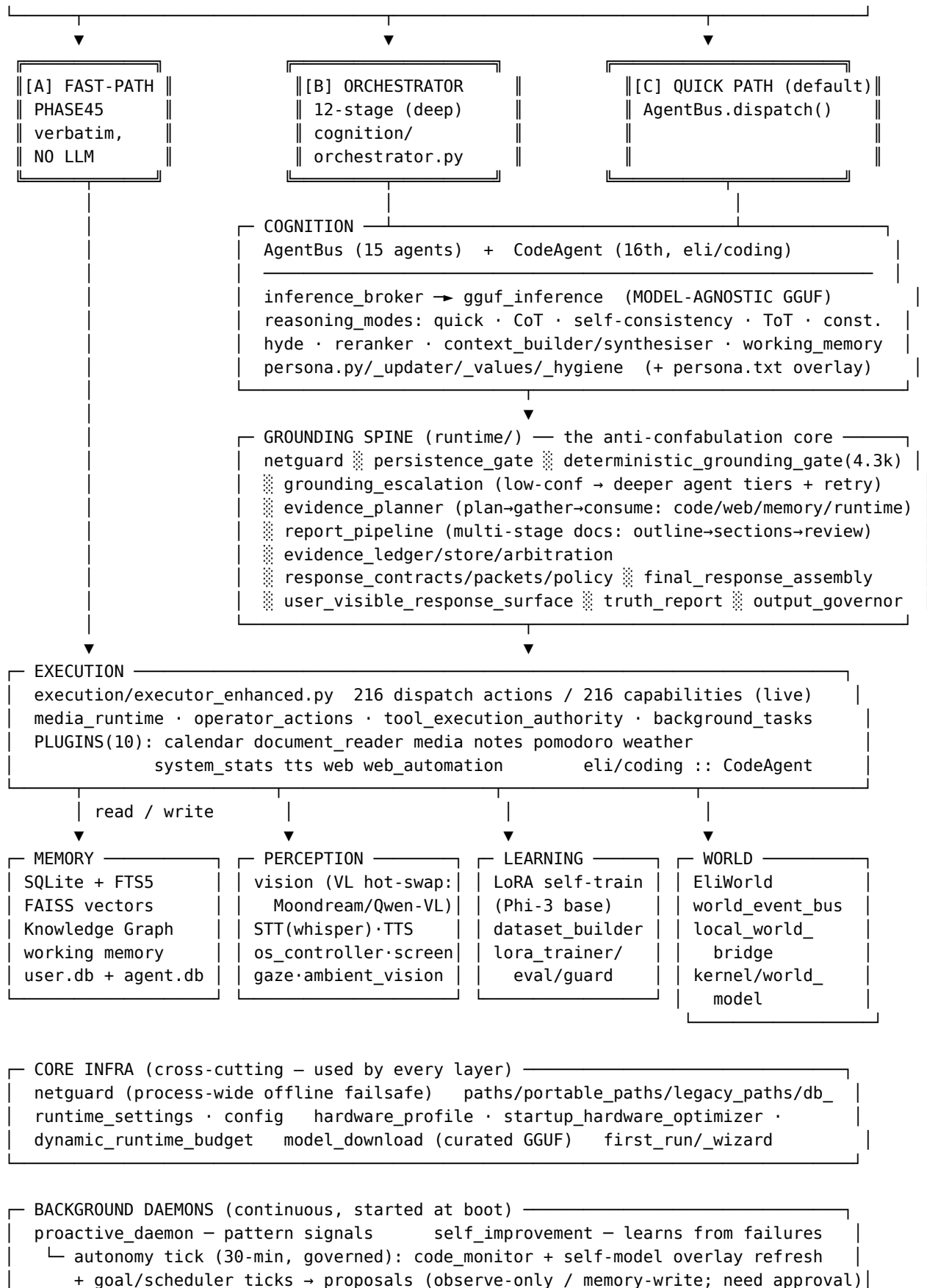
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## Blueprint — ELI MKXI Full Architecture (ASCII)

The entire system in one drawing, plus the module tree and data layout. Grounded in the real source (see `architecture.md` for prose, `diagrams.md` for the pipeline/memory/gating close-ups). Every layer and box maps to a real path.

### A. Full system — layered block diagram





|                                           |                                       |
|-------------------------------------------|---------------------------------------|
| habits_scheduler/habits – routines        | scheduler · task_bus – jobs           |
| background_tasks – async heavy work       | reflection loop   ambient_vision loop |
| scheduled_tasks (durable overnight/timed) | world_event_bus ← confidence events   |

OUTPUT ← output\_governor / sanitiser ← inference ← (any path)  
           |  
           └→ TTS (Piper) / GUI render

## B. The 15 agents

AGENT BUS (cognition/agent\_bus.py :: AgentBus.dispatch → DispatchResult)  
 selects a set → runs on a dependency DAG → aggregates grounding\_confidence

|                      |                                                                                                       |
|----------------------|-------------------------------------------------------------------------------------------------------|
| BusMemoryAgent       | recall: SQLite/FTS5/FAISS → context                                                                   |
| KnowledgeGraphAgent  | entities/relations from the KG                                                                        |
| SystemAgent          | runs executor actions (WEB_SEARCH, RUNTIME_AUDIT, ...)                                                |
| OrchestratorAgent    | plan / coordinate grounded synthesis                                                                  |
| FileCodeAgent        | searches the codebase for code-grounded answers                                                       |
| IntrospectionBus...  | live runtime + gathers identity/awareness audits as evidence (persona summarises – never a data dump) |
| CapabilityAgent      | capability manifest (what ELI can do)                                                                 |
| ReflectionAgent      | session reflection / insights                                                                         |
| ProactiveAgent       | proactive patterns / signals                                                                          |
| HabitAgent           | learned habits                                                                                        |
| SelfImprovementAgent | self-improvement state                                                                                |
| FrontierAgent        | frontier reasoning hooks                                                                              |
| PluginAgent          | dispatch to registered plugins                                                                        |
| VoiceAgent           | STT/TTS engine state                                                                                  |
| CriticAgent          | second-tier verifier: checks retriever agents' output                                                 |
| + CodeAgent (16th)   | SEPARATE pipeline: plan→search→verify→repair                                                          |
| eli/coding/agent     | (CODE_SOLVE / GENERATE_SCRIPT) – not a bus agent                                                      |

## C. Module tree (LOC · key files · role)

```

eli/ (~156k LOC, 392 files) · api/server.py (FastAPI web app + dashboard)
├── __main__.py ..... entry dispatch (GUI | --headless)
├── kernel/          13.1k — the core
│   ├── engine.py    12.0k   CognitiveEngine.process() – the spine *god-file
│   ├── scheduler.py          timed jobs
│   └── task_bus.py · pipeline.py · state.py · world_model.py · self_upgrade.py
├── execution/       20.5k — route → act
│   ├── executor_enhanced.py  15.0k  216 actions *god-file
│   ├── router_enhanced.py    6.1k   priority pipeline *god-file
│   ├── execution_planner.py · route_authority.py · route_contracts.py
│   ├── tool_execution_authority.py · operator_actions.py · operator_policy.py
│   └── media_runtime.py · router_plugin_intents.py · executor_plugin_handlers.py

```

- └─ portable\_intent\_contract.py · execution\_intent\_packets.py
- cognition/ 12.3k — think
  - └─ agent\_bus.py 2.4k 15 agents + dispatch \*
  - └─ orchestrator.py 12-stage deep retrieval
  - └─ gguf\_inference.py 2.1k · inference\_broker.py model-agnostic inference
  - └─ reasoning\_modes.py · hyde.py · reranker.py · llm\_intent.py
  - └─ persona.py/\_updater/\_values/\_status/\_hygiene (+ persona.txt, persona.auto.txt)
  - └─ context\_builder.py · context\_synthesiser.py · working\_memory.py
  - └─ output\_governor.py · response\_governance.py · response\_sanitizer.py
  - └─ grounded\_status.py · introspection\_agent.py · tone\_analyzer.py · chat\_model.py
- runtime/ 18.0k — grounding spine + daemons (68 files)
  - └─ deterministic\_grounding\_gate.py 4.3k \* · grounding\_escalation.py
  - └─ evidence\_ledger/store/arbitration.py · memory\_evidence.py
  - └─ persistence\_gate.py · truth\_report.py · control\_contracts.py
  - └─ response\_contracts/packets/policy.py · final\_response\_assembly/provider.py
  - └─ user\_visible\_response\_surface.py · personal\_memory\_\*.py · reflection.py
  - └─ background\_tasks.py · self\_improvement.py · code\_monitor.py
  - └─ capability\_sync.py · pending\_proposal.py · runtime\_policy.py
- memory/ 6.6k — remember (13 files)
  - └─ memory.py 4.5k Memory · SQLite + FTS5 \*god-file
  - └─ vector\_store.py FAISS index
- perception/ 5.4k — sense (18 files)
  - └─ vision.py · analyze\_image/csv/pdfs/mesh.py · ambient\_vision.py
  - └─ audio\_stt.py · local\_whisper\_stt.py · voice\_worker(\_streaming).py · eli\_listen.py
  - └─ tts\_router.py · os\_controller.py · screen\_locator.py · gaze\_engine.py
- planning/ 3.1k — proactivity (21 files)
  - └─ proactive\_daemon.py · habits\_scheduler.py · habits.py · jobqueue\_cli.py
- coding/ 1.5k — CodeAgent (plan→search→verify→repair, 9 files)
- learning/ 3.1k — LoRA self-training (11 files)
  - └─ lora\_trainer/eval/guard.py · dataset\_builder/filters.py
  - └─ bootstrap\_phi3\_base.py · base\_model\_resolver.py · training\_preflight.py
- plugins/ 1.9k — manager + 10 plugins
  - └─ calendar · document\_reader · media · notes · pomodoro · weather · system\_stats · tts · web · web\_automation
- world/ 1.5k — EliWorld (world\_event\_bus, local\_world\_bridge)
- core/ 4.9k — infra
  - └─ netguard.py offline failsafe + allow\_network()
  - └─ paths.py · portable\_paths.py · legacy\_paths.py · db\_paths.py
  - └─ runtime\_settings.py · config.py · grounding.py
  - └─ hardware\_profile.py · startup\_hardware\_optimizer.py · dynamic\_runtime\_budget.py
  - └─ model\_download.py · first\_run.py · first\_run\_wizard.py
- gui/ 18.7k — PySide6 desktop
  - └─ eli\_pro\_audio\_gui\_v2\_0.py 11.0k \*god-file · app.py · labs\_tab.py 5.7k
  - └─ panels/ (startup.py: model picker + FirstBootWizard, HardwareTuningDock)

```

|
|— tools/          8.1k — image_engine · news · document tools
|— contracts/ 0.7k · cli/ 0.1k · system/ 0.3k · utils/ 0.9k

```

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## D. On-disk data layout

```

artifacts/
|— db/
|   |— user.sqlite3      memories(+_fts) · conversation_turns · conversations ·
|   |                   kg_entities(+_fts) · kg_relations · recall_log ·
|   |                   runtime_events · news_articles(+_fts) · news_reflections ·
|   |                   habits/habit_events/habit_rules · observations ·
|   |                   learning_replay · working_memory_pins · user_patterns ·
|   |                   session_summaries · corrections · failures
|   |— agent.sqlite3    agent_dispatches · agent_metrics · improvements ·
|   |                   failures · code_patches · error_tracking · observations
|— vectors/index.faiss  semantic index (embedder: nomic-embed-...Q4_K_M.gguf)
|— conversations/ + archive/
|— runtime/users/<uuid>/user_profile...  per-user profile
|— runtime_snapshot.json                live model/runtime truth
|— world/{snapshots,ledger,journal}/ · proactive/ · incidents/<date>.jsonl
|— documents/ · scripts/ · analyze_image_*/
|
config/  settings.json (gitignored) · templates/settings.template.json ·
        settings.example.json · plugins_state.json
models/  <your>.gguf · embeddings/ · whisper/ · image/          voices/ *.onnx (Piper)

```

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## E. The flow in one line

```

INPUT → ROUTER → ENGINE — FAST-PATH —————→ OUTPUT (verbatim)
                        |— ORCHESTRATOR → retrieval —┐
                        |— QUICK → AGENT BUS ———┐
MEMORY (SQL/FTS5/FAISS/KG/WM) → context ←——┘
GATES: netguard · persistence · grounding · escalation · confidence · governor
INFERENCE (broker → gguf, model-agnostic) → OUTPUT → TTS/GUI
BACKGROUND: proactive · self-improve · habits · learning(LoRA) · world · scheduler

Companion docs: architecture.md (prose, every subsystem) · diagrams.md (pipeline / memory
/ gating close-ups) · tools/eval/README.md (the measurement layer). ““

```